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Interconnection structure and manufacture method thereof

Abstract

An interconnection structure includes a first dielectric layer, a first conduction layer, a conductor pillar, an upper dielectric layer and an upper conduction layer. The first dielectric layer is disposed over a first terminal of a device. The first conduction layer is disposed over the first dielectric layer. The conductor pillar is connected to the first terminal. The upper dielectric layer is disposed over the first conduction layer. The upper conduction layer is disposed over the upper dielectric layer. The conductor pillar connects to the upper conduction layer but disconnects from the first conduction layer.

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Background/Summary

(1) This application claims the benefit of U.S. provisional application Ser. No. 63/158,896, filed Mar. 10, 2021, the subject matter of which is incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE

Field of the Disclosure

(1) The disclosure relates in general to a semiconductor structure and the manufacture method thereof, and more particularly to an interconnection structure and the manufacture method thereof.

Description of the Related Art

(2) A current Integrated-circuits chip has many transistors which are connected by a one-die interconnection system of using multiple interconnections, which has its first interconnection layer M1 to connect the gate-level (Gate) and the diffusion-level of the source-region and the drain-region (called generally as Diffusion) of the MOSFET device. When there is a need to increase a second interconnection layer M2 for facilitating signal transmission without enlarging the die size by only using M1, then a structure Via-1, which is composed of some types of the conductive materials, is formed for connecting M2 to M1. Thus, there is a vertical structure which is formed from the diffusion through a contact (Con) connection to M1, i.e. "Diffusion-Con-M1"; similarly another structure to connect the gate through a contact structure to M1 can be formed as "Gate-Con-M1". Additionally, if a connection structure is needed to be formed from an M1 interconnection through a Via1 to connect to an M2 interconnection, then it is named as "M1-Via1-M2". A more complex interconnection structure from the Gate-level to the M2 interconnection can be described as "Gate-Con-M1-Via1-M2". Furthermore, a stacked interconnection system may have an "M1-Via1-M2-Via2-M3-Via4-M4 . . ." structure.

(3) By so far, the state-of-the-art interconnection system may not allow, for example, that the gate directly connect to M2 without bypassing the M1 structure. As results, the necessary space between one M1 interconnection and the other M1 interconnection can increase the die size and in some cases the wiring connections may block some efficient channeling intention of using M2 directly to surpass M1 regions. In addition, there is no method to form a self-alignment structure between Via1 to contact and at the same time both Via1 and contact are connected to their own interconnection systems, respectively.

(4) Therefore, there is a need to provide an advanced interconnection structure and the manufacture method thereof to overcome the drawbacks of the prior art.

SUMMARY OF THE DISCLOSURE

(5) One embodiment of the present disclosure is to provide an interconnection structure, wherein the interconnection structure includes a first dielectric layer, a first conduction layer, a conductor

pillar, an upper dielectric layer and an upper conduction layer. The first dielectric layer is disposed over a first terminal of a device. The first conduction layer is disposed over the first dielectric layer. The conductor pillar is connected to the first terminal. The upper dielectric layer is disposed over the first conduction layer. The upper conduction layer is disposed over the upper dielectric layer. The conductor pillar connects to the upper conduction layer but disconnects from the first conduction layer.

(6) In one aspect of the present disclosure, the conductor pillar includes a first conductor pillar portion and a second conductor pillar portion, the first conductor pillar portion is surrounded by the first dielectric layer and the second conductor pillar portion is surrounded by the upper dielectric layer.

(7) In another aspect of the present disclosure, the first conductor pillar portion is formed based on a silicon region of the first terminal, and the second conductor pillar portion is formed based on the first conductor pillar portion.

(8) In another aspect of the present disclosure, the device is a transistor and the first terminal of the device is a gate terminal. The gate terminal includes a gate dielectric layer, a gate conduction layer over the gate dielectric layer, and the silicon region over the gate conduction layer.

(9) In another aspect of the present disclosure, the silicon region is a poly-silicon region or an amorphous silicon region.

(10) In another aspect of the present disclosure, the first conductor pillar portion is formed by a selective epitaxy growth based on the poly-silicon region or the amorphous silicon region, and the second conductor pillar portion is formed by a selective epitaxy growth based on the first conductor pillar portion.

(11) In another aspect of the present disclosure, the device is a transistor and the first terminal of the device is a drain terminal.

(12) In another aspect of the present disclosure, a top surface of the conductor pillar is higher than a top surface of the upper dielectric layer.

(13) Another embodiment of the present disclosure is to provide a manufacture method of an interconnection structure, wherein the manufacture method includes steps as follows: A first dielectric layer is formed over a first terminal of a device. A conductor pillar is formed to be connected to the first terminal. A first conduction layer is formed over the first dielectric layer. An upper dielectric layer is formed over the first conduction layer. An upper conduction layer is formed over the upper dielectric layer. Wherein the conductor pillar connects to the upper conduction layer but disconnects from the first conduction layer.

(14) In one aspect of the present disclosure, the conductor pillar comprises a first conductor pillar portion and a second conductor pillar portion, and the step of forming the conductor pillar includes steps as follows: Firstly, an open hole is formed in the first dielectric layer to reveal a silicon region of the first terminal. Next, the first conductor pillar portion is on the silicon region of the first terminal by a first selective epitaxy growth. The second conductor pillar portion is then formed on the first conductor pillar by a second selective epitaxy growth.

(15) In another aspect of the present disclosure, before the step of forming the second conductor pillar portion, the manufacture method further includes a step of forming a first dielectric sub-layer over the first dielectric layer, wherein a top surface of the first dielectric sub-layer has a level substantially the same as that of the first conductor pillar portion.

(16) In another aspect of the present disclosure, a width of the open hole is equal to a minimum feature size.

(17) In another aspect of the present disclosure, both the first conductor pillar portion and the second conductor pillar portion are highly doped silicon pillars.

(18) In another aspect of the present disclosure, the step of forming the first conduction layer includes steps as follows: Firstly, a first conduction material is deposited over the first dielectric layer. Next, a second dielectric sub-layer is deposited over the first conduction material. The first

conduction material and the second dielectric sub-layer are then patterned to form the first conduction layer and to define an opening hollow passing the first conduction layer and the second dielectric sub-layer; wherein the conductor pillar penetrates through the opening hollow.

(19) In another aspect of the present disclosure, the step of forming the upper dielectric layer includes steps as follows: Firstly, an upper dielectric material is disposed to cover the first dielectric sub-layer and fill in the opening hollow; and the upper dielectric material is then etched back; such that, a top surface of the upper dielectric layer is lower than that of the conductor pillar.

(20) In another aspect of the present disclosure, a width of the opening hollow is greater than a minimum feature size.

(21) Yet another embodiment of the present disclosure is to provide an interconnection structure, wherein the interconnection structure includes a first dielectric layer, a first dielectric sub-layer, a conductor pillar and a first conduction layer. The first dielectric layer is disposed over a first terminal of a device. The first dielectric sub-layer is disposed over the first dielectric layer. The conductor pillar is connected to the first terminal. The first conduction layer is disposed over the first dielectric sub-layer and connected to the conductor pillar. The device is a transistor and the first terminal of the device is a gate terminal which includes a gate dielectric layer, a gate conduction layer over the gate dielectric layer, and a silicon region over the gate conduction layer, and the conductor pillar is connected to the silicon region of the gate terminal.

(22) In one aspect of the present disclosure, a top surface of the conductor pillar has a level substantially the same as that of the first dielectric sub-layer.

(23) In another aspect of the present disclosure, the conductor pillar includes a tungsten pillar and a TiN layer surrounding the tungsten pillar.

(24) In another aspect of the present disclosure, a width of the conductor pillar is equal to a minimum feature size.

(25) In another aspect of the present disclosure, the device further includes a second terminal serving as a drain terminal, and the interconnection structure further includes another conductor pillar and another first conduction layer. The another conductor pillar is connected to the drain terminal. The another first conduction layer is disposed over the first dielectric sub-layer and connected to the another conductor pillar.

(26) Yet another embodiment of the present disclosure is to provide a manufacture method of an interconnection structure, wherein the manufacture method includes steps as follows: A first dielectric layer is formed over a first terminal and a second terminal of a device. A first silicon pillar is formed passing through the first dielectric layer and connected to the first terminal, and a second silicon pillar is formed passing through the a first dielectric layer and connected to the second terminal, simultaneously.

(27) In another aspect of the present disclosure, both the first silicon pillar and the second silicon pillar includes highly doped silicon; and the method further includes steps of simultaneously forming a first connection sub-layer connected to the first silicon pillar, and forming a second connection sub-layer connected to the second silicon pillar.

(28) In another aspect of the present disclosure, the manufacture method further includes steps of replacing the first silicon pillar by a first conductor pillar and replacing the second silicon pillar by a second conductor pillar; wherein the first conductor pillar is connected to the first terminal and the second conductor pillar is connected to the second terminal.

(29) In another aspect of the present disclosure, the manufacture method further includes steps of simultaneously forming a first connection sub-layer connected to the first conductor pillar, and forming a second connection sub-layer connected to the second conductor pillar.

(30) In another aspect of the present disclosure, the manufacture method further includes steps of forming a first dielectric sub-layer over the first dielectric layer; and planarizing the first dielectric sub-layer, the first silicon pillar and the second silicon pillar, such that a top surface of the first dielectric sub-layer has a level substantially the same as that of the first silicon pillar and as that of

the second silicon pillar.

(31) In another aspect of the present disclosure, the step of forming the first and the second silicon pillars includes steps as follows: Firstly, open holes are formed in the first dielectric layer to reveal a silicon region of the first terminal and a silicon region of the second terminal. The first silicon pillar is then formed on the silicon region of the first terminal and the second silicon pillar is formed on the silicon region of the second terminal simultaneously by a selective epitaxy growth.

(32) Yet another embodiment of the present disclosure is to provide an interconnection structure, wherein the interconnection structure includes a first dielectric layer, a conductor pillar, a first highly doped silicon pillar and first conduction layer. The first dielectric layer is disposed over a first terminal of a device. The conductor pillar is connected to the first terminal. The first highly doped silicon pillar is disposed on the conductor pillar. The first conduction layer is disposed over the first dielectric layer and connected to the highly doped silicon pillar.

(33) In another aspect of the present disclosure, the interconnection structure further includes a first dielectric sub-layer over the first dielectric layer; wherein the first highly doped silicon pillar is surrounded by the first dielectric sub-layer.

(34) In another aspect of the present disclosure, the conductor pillar includes a tungsten pillar and a TiN layer surrounding the tungsten pillar.

(35) In another aspect of the present disclosure, the interconnection structure further includes a highly doped silicon side pillar on the first highly doped silicon pillar and connected to the first metal layer.

(36) Yet another embodiment of the present disclosure is to provide a manufacture method of an interconnection structure, wherein the manufacture method includes steps as follows: A first dielectric layer is formed over a first terminal of a device. Next, a silicon pillar is formed to be connected to the first terminal. The silicon pillar is then replaced by a conductor pillar, wherein the conductor pillar is connected to the first terminal. An upper portion of the conductor pillar is replaced by a first highly doped silicon pillar. Subsequently, a first conduction layer is formed to be connected to the first highly doped silicon pillar.

(37) In another aspect of the present disclosure, before the step of replacing the silicon pillar, the manufacture further includes steps as follows: Firstly, a first dielectric sub-layer is formed over the first dielectric layer. The first dielectric sub-layer and the silicon pillar are then patterned, such that a top surface of the first dielectric sub-layer has a level substantially the same as that of the silicon pillar.

(38) In another aspect of the present disclosure, the step of forming the silicon pillar includes steps of forming an open hole in the first dielectric layer to reveal a silicon region of the first terminal; and forming the silicon pillar on the silicon region of the first terminal by a selective epitaxy growth.

(39) In another aspect of the present disclosure, the conductor pillar includes a tungsten pillar and a TiN layer, and the step of replacing the silicon pillar by the conductor pillar includes steps as follows: Firstly, the silicon pillar is removed to reveal the open hole. Next the TiN layer is formed in the open hole; and the tungsten pillar is then formed to be surrounded by the TiN layer.

(40) In another aspect of the present disclosure, the manufacture method further includes a step of forming a side pillar on the first highly doped silicon pillar and connected to the first metal layer, wherein the side pillar includes a highly doped silicon material.

(41) Yet another embodiment of the present disclosure is to provide an interconnection structure, wherein the interconnection structure includes a first dielectric layer, a first conductor pillar, a first highly doped silicon pillar, a first conduction layer and a second conductor pillar. The first conductor pillar is surrounded by the first dielectric layer. The first highly doped silicon pillar is disposed on the first conductor pillar. The first conduction layer is disposed over the first dielectric layer. The second conductor pillar is disposed on the first highly doped silicon pillar and connected to the first conduction layer.

- (42) In another aspect of the present disclosure, the first conductor pillar, the first highly doped silicon pillar and the second conductor pillar are self-aligned in a vertical direction.
- (43) In another aspect of the present disclosure, each of the first conductor pillar and the second conductor pillar includes a tungsten pillar and a TiN layer surrounding the tungsten pillar.
- (44) In another aspect of the present disclosure, the interconnection structure further includes a first dielectric sub-layer over the first dielectric layer and below the first conduction layer, wherein the first highly doped silicon pillar is surrounded by the first dielectric sub-layer.
- (45) In another aspect of the present disclosure, the interconnection structure further includes a second highly doped silicon pillar on the second conductor pillar.
- (46) In another aspect of the present disclosure, the interconnection structure further includes a second dielectric sub-layer over the first conduction layer, wherein the second highly doped silicon pillar is surrounded by the second dielectric sub-layer.
- (47) Yet another embodiment of the present disclosure is to provide an interconnection structure, wherein the interconnection structure includes a lower conduction layer, a first conductor pillar, a first highly doped silicon pillar, a lower dielectric layer, an upper conduction layer and a second conductor pillar. The first conductor pillar is surrounded by and connected to the lower conduction layer. The first highly doped silicon pillar is disposed on the first conductor pillar. The lower dielectric layer is disposed over the lower connection layer. The upper conduction layer is disposed over the lower dielectric layer. The second conductor pillar is disposed on the first highly doped silicon pillar and surrounded by the upper conduction layer. The second conductor pillar is connected to the upper conduction layer.
- (48) In another aspect of the present disclosure, the first conductor pillar, the first highly doped silicon pillar and the second conductor pillar are self-aligned in a vertical direction.
- (49) In another aspect of the present disclosure, the interconnection structure further includes a lower dielectric sub-layer between the lower dielectric layer and the lower conduction layer, wherein the first highly doped silicon pillar is surrounded by the lower dielectric sub-layer.
- (50) In another aspect of the present disclosure, the interconnection structure further includes a second highly doped silicon pillar on the second conductor pillar.
- (51) In another aspect of the present disclosure, the interconnection structure further includes an upper dielectric sub-layer over the upper conduction layer, wherein the second highly doped silicon pillar is surrounded by the upper dielectric sub-layer.
- (52) Yet another embodiment of the present disclosure is to provide an interconnection structure, wherein the interconnection structure includes a first dielectric layer over a first conductive terminal; a first conduction layer over the first dielectric layer; a conductor pillar penetrating through the first dielectric layer and connected to the first conductive terminal; an upper dielectric layer over the first conduction layer; and an upper conduction layer over the upper dielectric layer. Wherein the conductor pillar penetrates through the upper dielectric layer and connects to the upper conduction layer, but the conductor pillar does not electrically connect the first conduction layer.
- (53) In another aspect of the present disclosure, the interconnection structure further comprises a plurality of middle conduction layers between the first conduction layer and the upper conduction layer, wherein each middle conduction layer vertically shifts with each other, and the conductor pillar does not electrically connect the plurality of middle conduction layers.
- (54) In another aspect of the present disclosure, the conductor pillar comprises a first conductor pillar portion and a second conductor pillar portion, the first conductor pillar portion is surrounded by the first dielectric layer and the second conductor pillar portion is surrounded by the upper dielectric layer.
- (55) In another aspect of the present disclosure, the first conductor pillar portion is formed based on a seed region of the first conductive terminal, and the second conductor pillar portion is formed based on a seed region of the first conductor pillar portion.
- (56) In another aspect of the present disclosure, the seed region of the first conductive terminal is

made of a poly-silicon or an amorphous silicon.

(57) In another aspect of the present disclosure, the first conductor pillar portion is formed by a selective epitaxy growth based on the poly-silicon or the amorphous silicon of the first conductive terminal.

(58) In another aspect of the present disclosure, the first conductive terminal is a gate terminal of a semiconductor transistor of a semiconductor substrate; wherein the gate terminal comprises a gate dielectric layer, a gate metal layer over the gate dielectric layer, and a silicon region over the gate metal layer.

(59) In another aspect of the present disclosure, the first conductive terminal is a gate terminal of a semiconductor transistor of a semiconductor substrate; wherein the gate terminal comprises a gate dielectric layer, and a gate poly-silicon layer over the gate dielectric layer.

(60) In another aspect of the present disclosure, wherein the first conductive terminal is a drain terminal or a source terminal of a semiconductor transistor.

(61) In another aspect of the present disclosure, the first conductive terminal is a lower conduction layer over a semiconductor transistor of a semiconductor substrate, the lower conduction layer comprises a metal region and a seed region electrically coupled to the metal region.

(62) In another aspect of the present disclosure, wherein the seed region of the lower conduction layer is made of highly doped silicon.

(63) Yet another embodiment of the present disclosure is to provide an interconnection structure, wherein the interconnection structure includes a semiconductor transistor formed based on a semiconductor substrate, the semiconductor transistor having a gate terminal and a drain terminal; a first conductor pillar over the semiconductor transistor and extending upward; a seed pillar on the first conductor pillar and electrically connected to the first conductor pillar; a second conductor pillar on the seed pillar and extending upward; and an upper conduction layer over the second conductor pillar and being electrically connected to the second conductor pillar. Wherein the seed pillar is clipped by the first conductor pillar and the second conductor pillar.

(64) In another aspect of the present disclosure, the seed pillar is made of a highly doped silicon.

(65) In another aspect of the present disclosure, the first conductor pillar is electrically connected to the gate terminal or the drain terminal of the transistor.

(66) In another aspect of the present disclosure, the interconnection structure further comprises a lower conduction layer above the semiconductor transistor and under the upper conduction layer; wherein the lower conduction layer is electrically isolated from the first conductor pillar and the second conductor pillar.

(67) In another aspect of the present disclosure, the interconnection structure further comprises a lower conduction layer above the semiconductor transistor and under the upper conduction layer; wherein the lower conduction layer is electrically connected to the first conductor pillar.

(68) In another aspect of the present disclosure, the interconnection structure further comprises a middle conduction layer between the lower conduction layer and the upper conduction layer; wherein the middle conduction layer is electrically isolated from the first conductor pillar and the second conductor pillar.

(69) Yet another embodiment of the present disclosure is to provide an interconnection structure, wherein the interconnection structure includes a first dielectric layer over a first conductive terminal; a conductor pillar penetrating through the first dielectric layer and connected to the first conductive terminal; a seed pillar being on and electrically connected to the conductor pillar; and a first conduction layer over the first dielectric layer and electrically connected to the seed pillar.

(70) In another aspect of the present disclosure, the interconnection structure further comprises a side pillar being on the seed pillar and electrically connected to the first conduction layer.

(71) In another aspect of the present disclosure, the seed pillar or the side pillar is made of highly doped silicon, the side pillar is selectively grown based on the seed pillar, and the first conduction layer is made of metal.

(72) In another aspect of the present disclosure, the first dielectric layer comprising a dielectric sub-layer; wherein the seed pillar is surrounded by the dielectric sub-layer, a top surface of the seed pillar is aligned with a top surface of the dielectric sub-layer, and a top surface of the side pillar is higher than the top surface of the dielectric sub-layer.

(73) In another aspect of the present disclosure, wherein the first conductive terminal is a lower conduction layer over a semiconductor transistor of a semiconductor substrate, the lower conduction layer comprises a metal region and a seed region electrically coupled to the metal region.

(74) In another aspect of the present disclosure, wherein the seed region of the lower conduction layer is made of highly doped silicon.

(75) In another aspect of the present disclosure, wherein the first conductive terminal is a gate terminal of a semiconductor transistor of a semiconductor substrate; wherein the gate terminal comprises a gate dielectric layer, a gate metal layer over the gate dielectric layer, and a seed region over the gate metal layer.

(76) In another aspect of the present disclosure, the seed region of the gate terminal is made of highly doped silicon.

(77) Yet another embodiment of the present disclosure is to provide an interconnection structure, wherein the interconnection structure includes a first dielectric layer over a gate terminal of a semiconductor transistor formed on a semiconductor substrate; a conductor pillar penetrating through the first dielectric layer and connected to the gate terminal; and a first conduction layer over the first dielectric layer and electrically connected to the conductor pillar; wherein the gate terminal comprises a gate dielectric layer and a seed region over the gate dielectric layer.

(78) In another aspect of the present disclosure, the conductor pillar is selectively grown based on the seed region of the gate terminal.

(79) In another aspect of the present disclosure, the conductor pillar is made of highly doped silicon.

(80) In another aspect of the present disclosure, the gate terminal further comprises a gate metal layer between the gate dielectric layer and the seed region.

(81) In another aspect of the present disclosure, the interconnection structure further comprises a seed pillar being on the conductor pillar, wherein the seed pillar is electrically connected to the conductor pillar and the first conduction layer.

(82) In another aspect of the present disclosure, the conductor pillar comprises a tungsten pillar and a TiN layer surrounding the tungsten pillar.

(83) Yet another embodiment of the present disclosure is to provide an interconnection structure, wherein the interconnection structure includes a semiconductor transistor formed based on a semiconductor substrate, the semiconductor transistor having a gate terminal and a drain terminal; a first conductor pillar over the semiconductor transistor and extending in an upward direction, wherein the first conductor pillar comprises a first seed region; and a second conductor pillar on the first conductor pillar and extending in the upward direction, wherein the second conductor pillar comprises a second seed region; wherein a bottom surface of the second conductor pillar is self-aligned with a top surface of the first conductor pillar.

(84) In another aspect of the present disclosure, the first conductor pillar is made of a highly doped silicon, and the first seed region is a top region of highly doped silicon.

(85) In another aspect of the present disclosure, the first conductor pillar further comprises a first tungsten pillar and a first TiN layer surrounding the first tungsten pillar, and the first seed region is made of highly doped silicon and is on the first tungsten pillar.

(86) In another aspect of the present disclosure, wherein the bottom surface of the second conductor pillar is self-aligned with a top surface of the first seed region.

(87) In another aspect of the present disclosure, the interconnection structure further comprises a third conductor pillar on the second conductor pillar and extending in the upward direction, the

third conductor pillar comprising a third seed region; wherein the second conductor pillar further comprises a second tungsten pillar and a second TiN layer surrounding the second tungsten pillar, and the second seed region is made of highly doped silicon and is on the second tungsten pillar; wherein a bottom surface of the third conductor pillar is self-aligned with a top surface of the second seed region.

(88) In another aspect of the present disclosure, the interconnection structure further comprises a lower conduction layer over the semiconductor transistor and an upper conduction layer over the lower conduction layer; wherein the first conductor pillar is electrically connected to the gate terminal or the drain terminal of the semiconductor transistor, the third conductor pillar is electrically connected to the upper conduction layer, and the lower conduction layer is electrically isolated from the first conductor pillar, the second conductor pillar, and the third conductor pillar.

(89) In another aspect of the present disclosure, the interconnection structure further comprises a lower conduction layer over the semiconductor transistor, an upper conduction layer over the lower conduction layer, and a middle conduction layer between the lower conduction layer and the upper conduction layer; wherein the first conductor pillar is electrically connected to the lower conduction layer, the third conductor pillar is electrically connected to the upper conduction layer, and the middle conduction layer is electrically isolated from the first conductor pillar, the second conductor pillar, and the third conductor pillar.

(90) According to the above, exemplified embodiments which significantly improve the connection structures between devices and interconnections to form an efficient wiring system on a die for completion of integrated circuits. The first exemplified embodiment allows either Gate or Diffusion (source/drain) areas to be directly connected to the M2 interconnection layer (without a transitional layer M1) in a self-alignment way through one vertical conductive plug being composed of contact-A and Via1-A which are respectively formed during the construction phases of making contact and Via1 in the other locations on the same die. This embodiment could also be applied to allows M1 interconnection or conduction layer to be directly connected to the MX interconnection layer (without a transitional conduction layers M2, M3, . . . MX-1) in a self-alignment way through one vertical conductive or conductor plug.

(91) Another exemplified embodiment is that a landing pad created by a vertical conductor pillar which connects either gate or diffusion region, respectively, to a planar surface which is effectively created for landing the metal M1 interconnection layer. This avoids making the contact connection material to go through a very rough surface topography inside the contact hole regions with a high aspect ratio of the depth to the opening-hole.

(92) Another exemplified embodiment is that the M1 can be connected to the contact in a self-alignment way with no need in mask or design to reserve an extra M1 border area between the M1 width and the contact width.

(93) Another exemplified embodiment is to make the Via1 exactly on top of the contact in a self-alignment way (similarly for Via2 on top of Via1 which can also be exactly self aligned mutually, and then similarly for Via3 on top of Via2, etc.), but all layers of M1, M2, etc. reserve their respective connection function so as to complete their individual interconnections but with the smallest footprint areas under Via1 and contact and the advantages are extendible for all via layers from the top to the bottom. This invention discloses a method to make self-alignments from the bottom interconnection layer to the upper interconnection layer to achieve vertical connections between them, which is different from the conventional alignment method from the upper interconnection layer to the bottom interconnection layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects of the disclosure will become better understood with regard to the following detailed description of the preferred but non-limiting embodiment(s). The following description is made with reference to the accompanying drawings:

(2) FIGS. 1A(1) to 1H(3) are a series top views and cross-sectional views illustrating the manufacture method for forming a device having a transistor structure according to one embodiment of the present disclosure.

(3) FIGS. 2A(1) to 2C(3) are a series top views and cross-sectional views illustrating the manufacture method for forming a device having a transistor structure according to another embodiment of the present disclosure.

(4) FIGS. 3A(1) to 3B(3) are a series top views and cross-sectional views illustrating the manufacture method for forming another device having a transistor structure according to yet another embodiment of the present disclosure.

(5) FIGS. 4A(1) to 4B(3) are a series top views and cross-sectional views illustrating the manufacture method for forming another device having a transistor structure according to yet another embodiment of the present disclosure.

(6) FIGS. 5(1) to 5(3) are a top view and cross-sectional views illustrating the manufacture method for forming another device having a transistor structure according to yet another embodiment of the present disclosure.

(7) FIGS. 6A(1) to 6C(3) are a series top views and cross-sectional views illustrating the manufacture method for forming another device having a transistor structure according to yet another embodiment of the present disclosure.

(8) FIGS. 7A(1) to 7C(3) are a series top views and cross-sectional views illustrating the manufacture method for forming another device having a transistor structure according to yet another embodiment of the present disclosure.

(9) FIG. 8 is a cross-sectional view illustrating a device having a transistor structure according to further another embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

(10) The present disclosure provides an interconnection structure and the manufacture method thereof. The above and other aspects of the disclosure will become better understood by the following detailed description of the preferred but non-limiting embodiment(s). The following description is made with reference to the accompanying drawings:

(11) Several embodiments of the present disclosure are disclosed below with reference to accompanying drawings. However, the structure and contents disclosed in the embodiments are for exemplary and explanatory purposes only, and the scope of protection of the present disclosure is not limited to the embodiments. It should be noted that the present disclosure does not illustrate all possible embodiments, and anyone skilled in the technology field of the disclosure will be able to make suitable modifications or changes based on the specification disclosed below to meet actual needs without breaching the spirit of the disclosure. The present disclosure is applicable to other implementations not disclosed in the specification.

(12) The following embodiments are described by forming an interconnection structure for a device **10**. In some embodiments of the present disclosure, the transistor structure **100** adopted as an example can be an NMOS transistor; a structure for a PMOS transistor can be derived similarly except which has oppositely doped or formed materials in contrast to those of the NMOS transistor.

Embodiment 1

(13) According to one embodiment of the present disclosure, the method for forming a device **10** having a transistor structure **100** includes steps as follows: Step **S11**: forming a first dielectric layer over a first terminal of a device. Step **S12**: forming a conductor pillar connected to the first terminal. Step **S121**: forming an open hole in the first dielectric layer to reveal a silicon region of

the first terminal. Step **S122**: forming a first conductor pillar portion on the silicon region of the first terminal by a first selective epitaxy growth. Step **S123**: forming a first dielectric sub-layer over the first dielectric layer, wherein a top surface of the first dielectric sub-layer has a level substantially the same as that of the first conductor pillar portion (such as planarized by CMP process or etching process). Step **S124**: forming a second conductor pillar portion on the first conductor pillar by a second selective epitaxy growth. Step **S13**: forming a first conduction layer over the first dielectric layer. Step **S131**: depositing a first conduction material over the first dielectric layer. Step **S132**: depositing a second dielectric sub-layer over the first conduction material. Step **S132**: patterning the first conduction material and the second dielectric sub-layer to form the first conduction layer and to define an opening hollow passing the first conduction layer and the second dielectric sub-layer, wherein the conductor pillar penetrates through the opening hollow. Step **S14**: forming at least one upper dielectric layer over the first conduction layer. Step **S141**: depositing an upper dielectric material to cover the first dielectric sub-layer and fill in the opening hollow. Step **S142**: etching back the upper dielectric material, such that a top surface of the upper dielectric layer is lower than that of the conductor pillar. Step **S15**: forming an upper conduction layer over the upper dielectric layer; wherein the conductor pillar connects to the upper conduction layer but disconnects from the first conduction layer.

(14) Referring to Step **S11**: forming a first dielectric layer **120** over a first conductive terminal (Gate, Drain, or Source) of a device **10**. FIG. **1A(1)** is a top view illustrating a structure after a first dielectric layer **120** is formed over a transistor structure **100** previously formed in a semiconductor substrate **101**, according to one embodiment of the present disclosure. FIG. **1A(2)** is a cross-sectional view taken along the cutting line **C1A1** as depicted in FIG. **1A(1)**. FIG. **1A(3)** is a cross-sectional view taken along the cutting line **C1A2** as depicted in FIG. **1A(1)**.

(15) In the present embodiment, the semiconductor substrate **101** includes a silicon layer, such as a poly-silicon layer or an amorphous silicon layer. As shown in FIGS. **1A(1)** to **1A(3)**, the transistor structure **100** is formed in an active area of the silicon layer defined by a shallow trench isolator (STI) **105**. The transistor structure **100** has a gate terminal **102** formed on the active area, a transistor channel region **103** formed in the active area beneath the gate terminal **102** and source/drain regions **104** formed in the active area and adjacent to the transistor channel region **103**.

(16) The gate terminal **102** comprises a gate dielectric layer **102a**, a gate conduction layer **102b** formed over the gate dielectric layer **102a** and a silicon region **102c** formed over the gate conduction layer **102b**. In some embodiments of the present disclosure, the gate dielectric layer **102a** can be made of oxide or low-k dielectric; the gate conduction layer **102b** can be made of metal (not limited to this regard); and the silicon region **102c** can be made of polysilicon or amorphous silicon. In some embodiment the gate terminal **102** further includes a capping layer (e.g. a nitride layer) over the top of the silicon region **102c** and further includes at least one spacer (e.g. including a nitride spacer **102s1** and a thermal oxide spacer **102s2**) over the sidewalls of the gate dielectric layer **102a**, the gate conduction layer **102b** and the silicon region **102c**. The top portion of the silicon region **102c** can serve as the revealed top portion **11**; and the revealed top portion **12** can be the drain terminal of the source/drain regions **104**.

(17) The first dielectric layer **120** is formed on the semiconductor substrate **101** at least covering the active area of the transistor structure **100** including the gate terminal **102** and the source/drain regions **104** as well as the STI **105**. In some embodiments of the present disclosure, the first dielectric layer **120** is made of oxide or low-k dielectric.

(18) Referring to Step **S12**: forming a conductor pillar (such as, a first conductor pillar **130A**) connected to the revealed top portion **11**. In the present embodiment, the forming of the conductor pillar (e.g. the first conductor pillar **130A**) includes steps as follows: Firstly, a plurality of open holes (such as the open holes **107a** and **107b** are formed in the first dielectric layer **120** to reveal the top portion of the silicon **102c** region (referring to Step **S121**).

(19) FIG. 1B(1) is a top view illustrating a structure after the open holes **107a** and **107b** are formed in the first dielectric layer **120** according to one embodiment of the present disclosure. FIG. 1B(2) is a cross-sectional view taken along the cutting line C1B1 as depicted in FIG. 1B(1). FIG. 1B(2) is a cross-sectional view taken along the cutting line C1B2 as depicted in FIG. 1B(1).

(20) In some embodiments, the open holes **107a** and **107b** are formed by a photolithography process to remove portions of the first dielectric layer **120** to expose the portion the silicon region **102c** (serving as the revealed top portion or first terminal **11**) and expose the silicon region of the drain terminal of the source/drain regions **104** (serving as the revealed top portion or second terminal **12**). In one embodiment, the first terminal **11** and a second terminal **12** are the top region of the gate terminal and source/drain terminal, respectively. In one example, each of the open holes **107a** and **107b** could be a size equal to a minimum feature size (e.g. a critical size of the transistor structure **100** of the device **10**). Of course, the size of the open holes **107a** and **107b** could be larger than the minimum feature size.

(21) As shown in FIGS. 1B(1) to 1B(3) the gate (i.e. the gate terminal **102**) and the diffusion regions (i.e. the source/drain regions **104**) of the transistor structure **100** (e.g. MOSFET) are respectively exposed from the open holes **107a** and **107b** surrounded by the insulators (eg. the first dielectric layer **120**). The bottoms of the open holes **107a** and **107b** (i.e. the revealed top portion **11** and the revealed top portion **12**) are made of materials with either polycrystalline/amorphous silicon or crystalline silicon with heavily doped concentrations having high conductivity, respectively.

(22) How to form the plurality open holes **107a** and **107b** on top of the gate (i.e. the gate terminal **102**) and diffusion (i.e. the source/drain regions **104**) regions is already disclosed in U.S. application Ser. No. 17/468,683 filed on Sep. 8, 2021 (Title: TRANSISTOR STRUCTURE WITH METAL INTERCONNECTION DIRECTLY CONNECTING GATE AND DRAIN/SOURCE REGIONS) by the same inventor in the present invention. All contents of the aforesaid US application are incorporated by reference herein.

(23) Referring to Step S122: a first conductor pillar portion (or a first sub conductor pillar) **131a** is formed on the silicon region **102c** of the revealed top portion **11** (that is exposed from the open hole **107a**) by a selective epitaxy growth technique (SEG), and a third conductor pillar portion (or a third sub conductor pillar) **131b** is formed on the silicon region of the revealed top portion **12** (the drain terminal of the source/drain regions **104** that is exposed from the open hole **107b** by the same SEG, simultaneously. FIG. 1C(1) is a top view illustrating a structure after the first conductor pillar portion **131a** and the third conductor pillar portion **131b** are respectively formed in the open holes **107a** and **107b** according to one embodiment of the present disclosure. FIG. 1C(2) is a cross-sectional view taken along the cutting line C1C1 as depicted in FIG. 1C(1). FIG. 1C(2) is a cross-sectional view taken along the cutting line C1C2 as depicted in FIG. 1C(1). The exposed silicon region **102c** of the gate terminal and the exposed silicon region of the source/drain terminal are seed regions for the selective epitaxy growth technique (SEG) to grow pillars based on the seed regions.

(24) In the present embodiment, a SEG (or selective atomic layer deposition technique) is used to grow heavily doped conductive silicon plugs (or the conductor pillars) based on the revealed top portion **11** and the revealed top portion **12**, to form the first conductor pillar portion **131a** and the third conductor pillar portion **131b**. Wherein each of the first conductor pillar portion **131a** and the third conductor pillar portion **131b** has a top **131t** higher than that of the first dielectric layer **120**.

(25) A first dielectric sub-layer **140** is then formed over the first dielectric layer **120** to make the top surface **140s** of the first dielectric sub-layer **140** substantially coplanar with the top **131t** of the first conductor pillar portions **131a** and the third conductor pillar portion **131b** (referring to Step S123). FIG. 1D(1) is a top view illustrating a structure after the first dielectric sub-layer **140** is formed over the first dielectric layer **120** according to one embodiment of the present disclosure. FIG. 1D(2) is a cross-sectional view taken along the cutting line C1D1 as depicted in FIG. 1D(1).

FIG. 1D(3) is a cross-sectional view taken along the cutting line C1D2 as depicted in FIG. 1C(1). Wherein, the top surface **140s** of the formed first dielectric sub-layer **140** has a level substantially the same as that of the top **131t** of the first conductor pillar portions **131a** and the third conductor pillar portion **131b**.

(26) In some embodiments, the forming of the first dielectric sub-layer **140** can include steps as follows: A dielectric material (not shown), such as silicon oxide, silicon nitride, low-k dielectric or other suitable material is deposited on the first dielectric layer **120** and covering the top surface **131t** of the first conductor pillar portion **131a** and the third conductor pillar portion **131b**. A planarizing process, such as a chemical mechanical polishing (CMP) process or an etch back process, using the top surface **131t** of the first conductor pillar portions **131a** and the third conductor pillar portion **131b** as a stop layer is performed to remove a portion of the dielectric material, so as to expose the top surface **131t** of the first conductor pillar portion **131a** and the third conductor pillar portion **131b**. Those “Exposed Heads” (the exposed top surfaces **131t** of the first conductor pillar portion **131a** and the third conductor pillar portion **131b**) of the first conductor pillar portions **131a** and the third conductor pillar portion **131b** can be used as landing pads for the subsequent process for forming the interconnection structure. Furthermore, each of the first conductor pillar portions **131a** and the third conductor pillar portion **131b** has a seed region or seed pillar in the upper portion thereof, and such seed region or seed pillar could be used for the following selective epitaxy growth.

(27) Subsequently, a second conductor pillar portion (or a second sub conductor pillar) **132a** is formed on the first conductor pillar **131a** by a second selective epitaxy growth (referring to Step S123); and a fourth conductor pillar portion (or a fourth sub conductor pillar) **132b** is formed on the third conductor pillar portion **131b**, simultaneously. FIG. 1E(1) is a top view illustrating a structure after the second conductor pillar portion **132a** and the fourth conductor pillar portion **132b** are formed on the first conductor pillar portions **131a** and the third conductor pillar portion **131b**, according to one embodiment of the present disclosure. FIG. 1E(2) is a cross-sectional view taken along the cutting line C1E1 as depicted in FIG. 1E(1). FIG. 1E(3) is a cross-sectional view taken along the cutting line C1E2 as depicted in FIG. 1E(1).

(28) In some embodiments of the present disclosure, the process for forming the second conductor pillar portion **132a** and the fourth conductor pillar portion **132b** can be (but not limited to) identical to that for forming the first conductor pillar portion **131a** and the third conductor pillar portion **131b**. In the present embodiment, the second conductor pillar portion **132a** and the fourth conductor pillar portion **132b** can be heavily doped conductive silicon plugs respectively grown from the top surface **131t** of the first conductor pillar portions **131a** and the third conductor pillar portion **131b** by SEG and extending upwards beyond the top surface **140s** of the first dielectric sub-layer **140**. The second conductor pillar portion **132a** and the fourth conductor pillar portion **132b** have a dopant concentration different from (or the same to) that of the first conductor pillar portions **131a** and the third conductor pillar portion **131b**. The first conductor pillar portions **131a** and the second conductor pillar portion **132a** together form the first conductor pillar **130A**; and third conductor pillar portion **131b** and the fourth conductor pillar portion **132b** together form the second conductor pillar **130B**.

(29) Referring to Step S13: forming a first conduction layer **150** over the first dielectric layer. FIG. 1F(1) is a top view illustrating a structure after the first conduction layer **150** is formed over the first dielectric layer **120** according to one embodiment of the present disclosure. FIG. 1F(2) is a cross-sectional view taken along the cutting line C1F1 as depicted in FIG. 1F(1). FIG. 1F(3) is a cross-sectional view taken along the cutting line C1F2 as depicted in FIG. 1F(1).

(30) In the present embodiment, the forming of the first conduction layer **150** includes steps as follows: A first conduction material (not shown), such as copper (Cu), Aluminum (Al), tungsten (W) or other suitable conductive material, can be deposited on the top surface **140s** of the first dielectric sub-layer **140** over the first dielectric layer (referring to Step S131). A second dielectric sub-layer

160 is then deposited on the first conduction material (referring to Step **S132**). The first conduction material and the second dielectric sub-layer **160** are patterned to form the first conduction layer **150** and to define an opening hollow **109** passing the first conduction layer **150** and the second dielectric sub-layer **160**, wherein the first conductor pillar **130A** penetrates through the opening hollow **109** (referring to Step **S133**) without contacting the first conduction layer **150** and the second dielectric sub-layer **160**.

(31) In some embodiments of the present disclosure, the first conduction layer **150** and the second dielectric sub-layer **160** can be formed prior to the forming of the second conductor pillar portion **132a** and the fourth conductor pillar portion **132b**. In this case, the opening hollow **109** can be defined by an etching process passing the first conduction layer **150** and the second dielectric sub-layer **160** to expose the tops **131t** of the first conductor pillar portions **131a** and the third conductor pillar portion **131b**, prior to the forming of the second conductor pillar portion **132a** and the fourth conductor pillar portion **132b**.

(32) Referring to Step **S14**: forming at least one upper dielectric layer **170** over the first conduction layer **150**. FIG. **1G(1)** is a top view illustrating a structure after the upper dielectric layer **170** is formed over the over the first conduction layer **150** according to one embodiment of the present disclosure. FIG. **1G(2)** is a cross-sectional view taken along the cutting line **C1G1** as depicted in FIG. **1G(1)**. FIG. **1G(3)** is a cross-sectional view taken along the cutting line **C1G2** as depicted in FIG. **1G(1)**.

(33) In the present embodiment, the forming of the upper dielectric layer **170** includes steps as follows: Firstly, at least one layer of upper dielectric material (not shown) is deposited to cover the second dielectric sub-layer **160** and the first dielectric sub-layer **140** (the portion of the top surface **140s** exposed from the opening hollow **109**) to fill in the opening hollow **109** (referring to Step **S141**). The upper dielectric material is then etched back, such that a top surface **170s** of the upper dielectric layer **170** is lower than the top surface **130t** of the first conductor pillar **130A** and the second conductor pillar **130B** (referring to Step **S142**).

(34) Referring to Step **S15**: forming an upper conduction layer **180** over the upper dielectric layer **170**; wherein the first conductor pillar **130A** connects to the upper conduction layer **180** but disconnects from the first conduction layer **150**. FIG. **1H(1)** is a top view illustrating a structure after the conduction layer **180** is formed over the over the upper dielectric layer **170** according to one embodiment of the present disclosure. FIG. **1H(2)** is a cross-sectional view taken along the cutting line **C1H1** as depicted in FIG. **1H(1)**. FIG. **1H(3)** is a cross-sectional view taken along the cutting line **C1H2** as depicted in FIG. **1H(1)**.

(35) In some embodiments of the present disclosure, the process for forming of the upper conduction layer **180** can be identical to or different from that for forming the first conductive layer **150**.

(36) After series steps of down-stream process are performed, the forming of the device **10** can be implemented. As mentioned, each of the exposed silicon region **102c** of the gate terminal and the exposed silicon region of the source/drain terminal has seed regions for the selective epitaxy growth technique (SEG) to grow pillars based on the seed regions. Furthermore, each of the first conductor pillar portions **131a** and the third conductor pillar portion **131b** also has a seed region or seed pillar in the upper portion thereof, and such seed region or seed pillar could be used for the following selective epitaxy growth. This embodiment could also be applied to allows **M1** interconnection (a kind of conductive terminal) or conduction layer to be directly connected to the **MX** interconnection layer (without a transitional conduction layers **M2**, **M3**, . . . **MX-1**) in a self-alignment way through one vertical conductive or conductor plug, as long as there is a seed portion or seed pillar on the upper portion of the conductive terminal and the conductor pillar portions configured for following selective epitaxy growth technique. The seed portion or seed pillar is not limited to silicon, and any material which could be used as a seed configured for following selective epitaxy growth is acceptable.

(37) Furthermore, using FIG. 1E(3) as an example, the second conductor pillar portion **132a** is grown upward by SEG method based on the seed region or seed pillar of the first conductor pillar portions **131a**. Since the area of the seed region is the area of the top surface of the first conductor pillar portion **131a**, the area of the bottom surface of the second conductor pillar portion **132a** will be the same (or substantially the same) as that of the top surface of the first conductor pillar portions **131a**, and self-alignment is easily accomplished between the first conductor pillar portions **131a** and the second conductor pillar portion **132a**.

Embodiment 2

(38) According to one embodiment of the present disclosure, the method for forming a device **20** having a transistor structure **200** includes steps as follows: Step S21: forming a first dielectric layer over a first terminal and a second terminal of a device. Step S22: simultaneously forming a first silicon pillar passing through the first dielectric layer and connected to the first terminal and forming a second silicon pillar passing through the first dielectric layer and connected to the second terminal. Step S221: forming open holes in the first dielectric layer to reveal a silicon region of the first terminal and a silicon region of the second terminal. Step S222: simultaneously forming the first silicon pillar on the silicon region of the first terminal and forming the second silicon pillar on the silicon region of the second terminal by a selective epitaxy growth. Step S23: forming a first conduction layer over the first dielectric sub-layer.

(39) Referring to Step S21: forming a first dielectric layer **120** over a first terminal **21** and a second terminal **22** of a device **20**. In one embodiment, the first terminal **21** and the second terminal **22** are the top region of the gate terminal and source/drain terminal, respectively. FIG. 2A(1) is a top view illustrating a structure after a first dielectric layer **220** is formed over a transistor structure **200** that is previously formed in a semiconductor substrate **201**, according to one embodiment of the present disclosure. FIG. 2A(2) is a cross-sectional view taken along the cutting line C2A1 as depicted in FIG. 2A(1). FIG. 2A(3) is a cross-sectional view taken along the cutting line C2A2 as depicted in FIG. 2A(1).

(40) In the present embodiment, the semiconductor substrate **201** includes a silicon layer, such as a poly-silicon layer or an amorphous silicon layer. As shown in FIGS. 2A(1) to 2A(3), the transistor structure **200** is formed in an active area of the silicon layer defined by a STI **205**. The transistor structure **200** has a gate terminal **202** formed on the active area, a transistor channel region **203** formed in the active area beneath the gate terminal **202** and source/drain regions **204** formed in the active area and adjacent to the transistor channel region **203**.

(41) The gate terminal **202** comprises a gate dielectric layer **202a**, a gate conduction layer **202b** formed over the gate dielectric layer **202a**, a silicon region **202c** formed over the gate conduction layer **202b**, a capping layer (e.g. a nitride layer) covering the top of the silicon region **202c** and at least one spacer (e.g. including a nitride spacer **202s1** and an thermal oxide spacer **202s2**) covering the sidewalls of the gate dielectric layer **202a**, the gate conduction layer **202b** and the silicon region **202c**. The top portion of the silicon region **202c** could serve as the first terminal **21**; and the second terminal **22** can be the drain terminal of the source/drain regions **204**.

(42) The first dielectric layer **220** is formed on the semiconductor substrate **201** at least covering the active area including the gate terminal **202** and the source/drain regions **204** as well as the STI **205**.

(43) Referring to Step S22: simultaneously forming a first silicon pillar **231a** passing through the first dielectric layer **220** and connected to the first terminal **21** and forming a second silicon pillar **231b** passing through the a first dielectric layer **220** and connected to the second terminal **22**. FIG. 2B(1) is a top view illustrating a structure after the first silicon pillar **231a** and the second silicon pillar **231b** are formed according to one embodiment of the present disclosure. FIG. 2B(2) is a cross-sectional view taken along the cutting line C2B1 as depicted in FIG. 2B(1). FIG. 2B(3) is a cross-sectional view taken along the cutting line C2B2 as depicted in FIG. 2B(1). Those silicon pillar could be made of highly doped silicon for higher conductivity.

- (44) In the present embodiment, the forming of the first silicon pillar **231a** and the second silicon pillar **231b** includes steps as follows: Firstly, a plurality of open holes (such as the open holes **207a** and **207b**) are formed in the first dielectric layer **220** to respectively reveal the top portion of the silicon region **202c** of the first terminal **21** and the drain terminal of the source/drain regions **204** serving as the second terminal **22** (referring to Step S221).
- (45) Next, referring to Step S222, the first silicon pillar **231a** is formed on the silicon region **202c** of the first terminal **21** that is exposed from the open hole **207a** by a SEG, and the second silicon pillar **231b** is formed on the silicon region of the second terminal **22** that is exposed from the open hole **207b** by the same SEG, simultaneously. Wherein, each of the first silicon pillar **231a** and the second silicon pillar **231b** has a top surface **231t** higher than that of the first dielectric layer **220**.
- (46) In some embodiments, a first dielectric sub-layer **240** is then formed over the first dielectric layer **220** to make the top surface **240s** of the first dielectric sub-layer **240** substantially coplanar with the top surface **203t** of the first silicon pillar **231a** and the second silicon pillar **231b**. planarizing the first dielectric sub-layer.
- (47) Referring to Step S23: forming a first conduction layer **250** over the first dielectric sub-layer **240**. FIG. 2C(1) is a top view illustrating a structure after the first conduction layer **250** is formed over the first dielectric sub-layer **240** according to one embodiment of the present disclosure. FIG. 2C(2) is a cross-sectional view taken along the cutting line C2C1 as depicted in FIG. 2C(1). FIG. 2C(3) is a cross-sectional view taken along the cutting line C2C2 as depicted in FIG. 2C(1).
- (48) In the present embodiment, the first conduction layer **250** is a patterned metal layer and includes a first metal sub-layer **250a** and a second metal sub-layer **250b** serving as the connecting wires respectively landing on the top surface **203t** of the first silicon pillar **231a** and the second silicon pillar **231b**.
- (49) The exposed heads (the top surface **230t** of the first conductor pillar **230A** and the second conductor pillar **230B**) can be used as the landing pads for facilitating M1 (the first conduction layer **250**) to connect with the conductor pillars (the first conductor pillar **230A** and the second conductor pillar **230B**) to touch on both the gate (the first terminal **21**) or the diffusion areas (the second terminal **22**), respectively. There is no need to make larger metal M1 pad which need to have a large size covering the openings of holes (the open holes **207a** and **207b**) since there is no worry of doing any etching steps on oxide or dielectric related to these contact holes.
- (50) After series steps of down-stream process are performed, the forming of the device **20** can be implemented.

Embodiment 3

- (51) According to one embodiment of the present disclosure, the method for forming a device **30** having a transistor structure **200** includes steps as follows: Step S31: forming a first dielectric layer over a first terminal and a second terminal of a device. Step S32: simultaneously forming a first silicon pillar passing through the a first dielectric layer and connected to the first terminal and forming a second silicon pillar passing through the a first dielectric layer and connected to the second terminal. Step S321: forming open holes in the first dielectric layer to reveal a silicon region of the first terminal and a silicon region of the second terminal. Step S322: simultaneously forming the first silicon pillar on the silicon region of the first terminal and forming the second silicon pillar on the silicon region of the second terminal by a selective epitaxy growth. Step S33: replacing the first silicon pillar by a first conductor pillar and replacing the second silicon pillar by a second conductor pillar. Step S34: forming a first conduction layer over the first dielectric sub-layer.
- (52) The structure of the device **30** and the manufacture method thereof are similar to that of device **20**. The difference between these two is that, in the device **30**, a first conductor pillar **330A** and a second conductor pillar **330B** is used to take the place of the first silicon pillar **231a** and the second silicon pillar **231b**. Since the steps S31 to S32 for forming the device **30** are identical to the steps S21 to S22 for forming the device **20**, thus the detail procedural and material applied thereby will

not redundantly described here, and the identical elements of the embodiments are designated with the same reference numerals. The process for forming the device **30** is described from step **S33**.

(53) Referring to Step **S33**: replacing the first silicon pillar **231a** by a first conductor pillar **330A** and replacing the second silicon pillar **231b** by a second conductor pillar **330B**. FIG. **3A(1)** is a top view illustrating a structure after the first conductor pillar **330A** and the second conductor pillar **330B** are formed according to one embodiment of the present disclosure. FIG. **3A(2)** is a cross-sectional view taken along the cutting line **C3A1** as depicted in FIG. **3A(1)**. FIG. **3A(3)** is a cross-sectional view taken along the cutting line **C3A2** as depicted in FIG. **3A(1)**.

(54) In the present embodiment, the replacement of the first silicon pillar **231a** and the second silicon pillar **231b** includes steps as follows: Firstly, a selective etching technique is performed to remove the first silicon pillar **231a** and the second silicon pillar **231b** to make the silicon region **202c** of the first terminal **21** and the drain terminal of the source/drain regions **204** serving as the second terminal **22** respectively exposed from the open holes **207a** and **207b**.

(55) A metal-plug process is then performed to form a titanium nitride (TiN) layer **330n** on the sidewalls and the bottoms of the open holes **207a** and **207b**. Next, tungsten metal is deposited on the top surface **240s** of the first dielectric sub-layer **240** to fill the open holes **207a** and **207b**. Subsequently, a CMP process or an etching back process is performed to remove the portion of the tungsten metal disposed on the top surface **240s** of the first dielectric sub-layer **240**, thereby two tungsten pillars **330w** surrounded by the TiN layer **330n** are respectively formed in the open holes **207a** and **207b**.

(56) Wherein, the tungsten pillars **330w** and the portion of the TiN layer **330n** that are both disposed in the open hole **207a** together form the first conductor pillar **330A**. The tungsten pillars **330w** and the portion of the TiN layer **330n** that are both disposed in the open hole **207b** together form the second conductor pillar **330B**. Each of the first conductor pillar **330A** and the second conductor pillar **330B** has a top surface **330t** substantially coplanar with the top surface **240s** of the first dielectric sub-layer **240**.

(57) Referring to Step **S34**: forming a first conduction layer **350** over the first dielectric sub-layer **240**. FIG. **3B(1)** is a top view illustrating a structure after the first conduction layer **350** is formed over the first dielectric sub-layer **240** according to one embodiment of the present disclosure. FIG. **3B(2)** is a cross-sectional view taken along the cutting line **C3B1** as depicted in FIG. **3B(1)**. FIG. **3B(3)** is a cross-sectional view taken along the cutting line **C3B2** as depicted in FIG. **3B(1)**.

(58) In the present embodiment, the first conduction layer **350** is a patterned metal layer and includes a first metal sub-layer **350a** and a second metal sub-layer **350b** serving as the connecting wires respectively landing on the top surface **330t** of the first conductor pillar **330A** and the second conductor pillar **330B**.

(59) After series steps of down-stream process are performed, the forming of the device **30** can be implemented.

Embodiment 4

(60) According to one embodiment of the present disclosure, the method for forming a device **40** having a transistor structure **200** includes steps as follows: Step **S41**: forming a first dielectric layer over a first terminal and a second terminal of a device. Step **S42**: simultaneously forming a first silicon pillar passing through the a first dielectric layer and connected to the first terminal and forming a second silicon pillar passing through the a first dielectric layer and connected to the second terminal. Step **S43**: replacing the first silicon pillar by a first conductor pillar and replacing the second silicon pillar by a second conductor pillar. Step **S44**: simultaneously forming a first highly doped silicon pillar on the first conductor pillar and forming a second highly doped silicon pillar on the second conductor pillar. Step **S45**: forming a first conduction layer over the first dielectric sub-layer.

(61) The structure of the device **40** and the manufacture method thereof are similar to that of device **30**. The difference between these two is that the device **40** further includes a first highly doped

silicon pillar **410a** and a second highly doped silicon pillar **410b**. Since the steps **S41** to **S43** for forming the device **40** are identical to the steps **S31** to **S33** for forming the device **30**, thus the detail procedural and material applied thereby will not redundantly described here, and the identical elements of the embodiments are designated with the same reference numerals. The process for forming the device **40** is described from step **S44**.

(62) Referring to Step **S44**: simultaneously forming a first highly doped silicon pillar **410a** connected to the first conductor pillar **330A** and forming a second highly doped silicon pillar **410b** connected to the second conductor pillar **330B**. FIG. **4A(1)** is a top view illustrating a structure after the first highly doped silicon pillar **410a** and the second highly doped silicon pillar **410b** are formed over the first conductor pillar **330A** and the second conductor pillar **330B** as depicted in FIG. **4A(1)** according to one embodiment of the present disclosure. FIG. **4A(2)** is a cross-sectional view taken along the cutting line **C4A1** as depicted in FIG. **4A(1)**. FIG. **4A(3)** is a cross-sectional view taken along the cutting line **C4A2** as depicted in FIG. **4A(1)**.

(63) The forming of the first highly doped silicon pillar **410a** and the second highly doped silicon pillar **410b** includes steps as follows: Upper portions of the two tungsten pillars **330w** and the TiN layer **330n** respectively formed in the open holes **207a** and **407b** are removed by an etching process (such as, a selective etching technique). Subsequently, highly doped N⁺ poly silicon is deposited on the first dielectric sub-layer **240** to fill the portion of open holes **207a** and **207b** originally occupied by the removed upper portions of the two tungsten pillars **330w** and the TiN layer **330n**.

(64) A CMP process or an etching back process is then performed to remove the portion of the highly doped N⁺ poly silicon disposed on the top surface **240s** of the first dielectric sub-layer **240**, thereby the first highly doped silicon pillar **410a** and the second highly doped silicon pillar **410b** are formed in the upper portion of open holes **207a** and **207b** originally occupied by the removed upper portions of the tungsten pillars **330w** and the TiN layer **330n**. Wherein each of the first highly doped silicon pillar **410a** and the second highly doped silicon pillar **410b** is surrounded by the first dielectric sub-layer **240**, and has a top surface **410s** coplanaring with the top surface **240s** of the first dielectric sub-layer **240**. In an another way, a conductor pillar could include the tungsten pillars and the first highly doped silicon pillar, that is, the conductor pillar has a seed region or seed pillar in the upper portion thereof.

(65) Referring to Step **S45**: forming a first conduction layer **450** over the first dielectric sub-layer **240**. FIG. **4B(1)** is a top view illustrating a structure after the first conduction layer **450** is formed over the first dielectric sub-layer **240** according to one embodiment of the present disclosure. FIG. **4B(2)** is a cross-sectional view taken along the cutting line **C4B1** as depicted in FIG. **4B(1)**. FIG. **4B(3)** is a cross-sectional view taken along the cutting line **C4B2** as depicted in FIG. **4B(1)**.

(66) In the present embodiment, the first conduction layer **450** is a patterned metal layer and includes a first metal sub-layer **350a** and a second metal sub-layer **450b** serving as the connecting wires respectively landing on the top surface **410s** the first highly doped silicon pillar **410a** and the second highly doped silicon pillar **410b**. The exposed top surface **410s** the first highly doped silicon pillar **410a** and the second highly doped silicon pillar **410b** can be used as the landing pads for facilitating M1 (the first conduction layer **450**) to connect with the conductor pillars (the first conductor pillar **330A** and the second conductor pillar **330B**) to touch on both the gate (the first terminal **21**) or the diffusion areas (the second terminal **22**), respectively.

(67) After series steps of down-stream process are performed, the forming of the device **40** can be implemented.

Embodiment 5

(68) According to one embodiment of the present disclosure, the method for forming a device **50** having a transistor structure **200** includes steps as follows: Step **S51**: forming a first dielectric layer over a first terminal and a second terminal of a device. Step **S52**: simultaneously forming a first silicon pillar passing through the first dielectric layer and connected to the first terminal and forming a second silicon pillar passing through the a first dielectric layer and connected to the

second terminal. Step S53: replacing the first silicon pillar by a first conductor pillar and replacing the second silicon pillar by a second conductor pillar. Step S54: simultaneously forming a first highly doped silicon pillar on the first conductor pillar and forming a second highly doped silicon pillar on the second conductor pillar. Step S55: forming a first conduction layer over the first dielectric sub-layer. Step S56: forming highly doped silicon side pillars respectively on the first highly doped silicon pillar and the second highly doped silicon pillar.

(69) The structure of the device **50** and the manufacture method thereof are similar to that of device **40**. The difference between these two is that the device **50** further included highly doped silicon side pillars **520** connected to the first conduction layer **550**. Since the steps S51 to S54 for forming the device **50** are identical to the steps S41 to S44 for forming the device **40**, thus the detail procedural and material applied thereby will not redundantly described here, and the identical elements of the embodiments are designated with the same reference numerals. The process for forming the device **50** is described from step S55.

(70) FIG. 5(1) is a top view illustrating a structure after the first conduction layer **550** and the highly doped silicon side pillars are formed according to one embodiment of the present disclosure. FIG. 5(2) is a cross-sectional view taken along the cutting line C51 as depicted in FIG. 5(1). FIG. 5(3) is a cross-sectional view taken along the cutting line C52 as depicted in FIG. 5(1).

(71) The forming of the first conduction layer **550** (referring to step S55) includes steps as follows: Firstly, a patterned metal layer including a first metal sub-layer **550a** and a second metal sub-layer **550b** is formed to make the first metal sub-layer **550a** and the second metal sub-layer **550b** respectively landing on the first highly doped silicon pillar **410a** and the second highly doped silicon pillar **410b**.

(72) Next, referring to step S56: a portion of the first metal sub-layer **550a** and a portion of the second metal sub-layer **550b** are removed (for example, by an etching process) to partially expose the top surface **410s** of the first highly doped silicon pillar **410a** and the second highly doped silicon pillar **410b**. Subsequently, a SEG is performed base on the exposed portions of the top surface **410s** to grow two highly doped silicon side pillars **520** respectively on the first highly doped silicon pillar **410a** and the second highly doped silicon pillar **410b**, wherein the two highly doped silicon side pillars **520** respectively connect to the (vertical) sidewalls of the etched first metal sub-layer **550a** and the second metal sub-layer **550b**.

(73) Based on this structure, even if the width of the metal conduction layer (such as, the first metal sub-layer **550a** or the second metal sub-layer **550b**) is the same as that of the underneath contact plug (which may be as small as minimum feature size), then the photolithographic masking Misalignment tolerance can cause that the metal conduction layer cannot fully cover the contact (as shown in FIGS. 5(2) and 5(3)), though there is no worry about the resistance between the metal conduction layer and contact may be too high due to shortages of contact areas. The invention here is that further using SEG to grow some extra highly doped silicon material (side pillars **520**) to attach the vertical walls of the metal conduction layer.

(74) After series steps of down-stream process are performed, the forming of the device **50** can be implemented.

Embodiment 6

(75) According to one embodiment of the present disclosure, the method for forming a device **60** having a transistor structure **200** includes steps as follows: Step S61: forming a first dielectric layer over a first terminal and/or a second terminal of a device. Step S62: forming a first silicon pillar passing through the a first dielectric layer and connected to the first terminal and/or forming a second silicon pillar passing through the a first dielectric layer and connected to the second terminal. Step S63: replacing the first silicon pillar by a first conductor pillar and replacing the second silicon pillar by a second conductor pillar. Step S64: forming a first highly doped silicon pillar on the first conductor pillar and forming a second highly doped silicon pillar on the second conductor pillar. Step S65: forming a first conduction layer over the first dielectric sub-layer. Step

S66: forming a third conductor pillar and a fourth conductor pillar respectively on the first highly doped silicon pillar and the second highly doped silicon pillar, so as to make the first conduction layer surrounds and connects to the third conductor pillar and the fourth conductor pillar. **Step S67:** simultaneously forming a third highly doped silicon pillar on the third conductor pillar and forming a fourth highly doped silicon pillar on the fourth conductor pillar.

(76) The structure of the device **60** and the manufacture method thereof are similar to that of device **40**. The difference between these two is that the interconnection structure of the device **60** can further extended. Since the steps **S61** to **S64** for forming the device **60** are identical to the steps **S41** to **S44** for forming the device **40**, thus the detail procedural and material applied thereby will not redundantly described here, and the identical elements of the embodiments are designated with the same reference numerals. The process for forming the device **50** is described from step **S65**.

(77) Referring to Step **S65**: forming a first conduction layer **650** over the first dielectric sub-layer **240**. FIG. **6A(1)** is a top view illustrating a structure after the first conduction layer **650** is formed over the first dielectric sub-layer **240** FIG. **6A(2)** is a cross-sectional view taken along the cutting line **C6A1** as depicted in FIG. **6A(1)**.

(78) Prior to the forming of the first conduction layer **650**, two undoped silicon pillars **610** are formed respectively on the top surface **410s** of the first highly doped silicon pillar **410a** and the second highly doped silicon pillar **410b**. In the present embodiment, a SEG is performed base on the top surface **410s** of the first highly doped silicon pillar **410a** and the second highly doped silicon pillar **410b** to grow two undoped silicon pillars **610** respectively, wherein each of the two undoped silicon pillars **610** has a top surface **610s** higher than the top surface **240s** of the first dielectric sub-layer **240**.

(79) Next, the first conduction layer **650** is formed on the first dielectric sub-layer **240** to surround and connect to these two undoped silicon pillars **610**. In the present embodiment the first conduction layer **650** has a top surface **650s** lower than the top surface **610s** of these two undoped silicon pillars **610**. Subsequently, a dielectric sub-layer **640** is formed on the top surface **650s** of the first conduction layer **650** to surround and connect to these two undoped silicon pillars **610**. Wherein, the top surface **640s** of the second dielectric sub-layer **640** substantially coplanar with the top surface **610s** of these two undoped silicon pillars **610**.

(80) Referring to Step **S66**: forming a third conductor pillar **630A** and a fourth conductor pillar **630B** respectively on the first highly doped silicon pillar **410a** and the second highly doped silicon pillar **410b**, so as to make the first conduction layer **650** surrounding and connecting to the third conductor pillar **630A** and the fourth conductor pillar **630B**. FIG. **6B(1)** is a top view illustrating a structure after the third conductor pillar **630A** and the fourth conductor pillar **630B** are formed on the first highly doped silicon pillar **410a** and the second highly doped silicon pillar **410b** according to one embodiment of the present disclosure. FIG. **6B(2)** is a cross-sectional view taken along the cutting line **C6B1** as depicted in FIG. **6B(1)**. FIG. **6B(3)** is a cross-sectional view taken along the cutting line **C6B2** as depicted in FIG. **6B(1)**.

(81) In the present embodiment, the forming of the third conductor pillar **630A** and a fourth conductor pillar **630B** includes steps as follows: Firstly, these two undoped silicon pillars **610** are removed to form two open holes **607a** and **607b**, and a metal-plug process is then performed to form a TiN layer **630n** on the sidewalls and the bottoms of the open holes **607a** and **607b**. Next tungsten metal is deposited on the top surface **640s** of the second dielectric sub-layer **640** to fill the open holes **607a** and **607b**. Subsequently, a CMP process or an etching back process is performed to remove the portion of the tungsten metal disposed on the top surface **640s** of the second dielectric sub-layer **640**, thereby two tungsten pillars **630w** surrounded by the TiN layer **630n** are respectively formed in the open holes **607a** and **607b**.

(82) In another embodiment, besides the two undoped silicon pillars **610** are removed, the first highly doped silicon pillar **410a** and the second highly doped silicon pillar **410b** could be removed as well. Then a metal-plug process is performed to form a TiN layer **630n** and the tungsten pillar

630w in the open holes **607a** and **607b**. Therefore, the fourth conductor pillar **630B** will contact the second conductor pillar **330B**, and the third conductor pillar **630A** will contact the first conductor pillar **330A**, as shown in FIG. **6B(4)** and FIG. **6B()**. Thus, the resistance of this pillar structure could be lower.

(83) Wherein, the tungsten pillars **630w** and the portion of the TiN layer **630n** that are both disposed in the open hole **607a** together form the third conductor pillar **630A**. The tungsten pillars **630w** and the portion of the TiN layer **630n** that are both disposed in the open hole **607b** together form the fourth conductor pillar **630B**. Each of the third conductor pillar **630A** and the fourth conductor pillar **630B** has a top surface **630s** substantially coplanar with the top surface **640s** of the second dielectric sub-layer **640**.

(84) Referring to Step **S67**: simultaneously forming a third highly doped silicon pillar **660a** on the third conductor pillar **630A** and forming a fourth highly doped silicon pillar **660b** on the fourth conductor pillar **630B**. FIG. **6C(1)** is a top view illustrating a structure after the third highly doped silicon pillar **660a** and the fourth highly doped silicon pillar **660b** are formed over the third conductor pillar **630A** and the fourth conductor pillar **630B** according to one embodiment of the present disclosure. FIG. **6C(2)** is a cross-sectional view taken along the cutting line **C6C1** as depicted in FIG. **6C(1)**. FIG. **6C(3)** is a cross-sectional view taken along the cutting line **C6C2** as depicted in FIG. **6C(1)**.

(85) The forming of the third highly doped silicon pillar **660a** and the fourth highly doped silicon pillar **660b** includes steps as follows: Upper portions of the two tungsten pillars **630w** and the TiN layer **630n** respectively formed in the open holes **607a** and **607b** are removed by an etching process. Subsequently, highly doped N⁺ poly silicon is deposited on the second dielectric sub-layer **640** to fill the portion of open holes **607a** and **607b** originally occupied by the removed upper portions of the two tungsten pillars **630w** and the TiN layer **630n**.

(86) A CMP process or an etching back process is then performed to remove the portion of the highly doped N⁺ poly silicon disposed on the top surface **640s** of the second dielectric sub-layer **640**, thereby the third highly doped silicon pillar **660a** and the fourth highly doped silicon pillar **660b** are then formed in the upper portion of open holes **607a** and **607b**, wherein the third highly doped silicon pillar **660a** and the fourth highly doped silicon pillar **660b** are surrounded by the second dielectric sub-layer **640**, and has a top surface **660s** coplanar with the top surface **640s** of the second dielectric sub-layer **640**.

(87) Thereinafter, the first conduction layer **650** and the second dielectric sub-layer **640** are patterned to form a first metal sub-layer **650a** and a second metal sub-layer **650b** serving as the connecting wires respectively surround and connect to the third conductor pillar **630A** and the fourth conductor pillar **630B**.

(88) After series steps of down-stream process are performed, the forming of the device **60** can be implemented. Furthermore, using FIG. **6A(3)** and FIG. **6C(3)** as an example, the undoped silicon pillar **610** is first grown upward by SEG method based on the seed region of the first highly doped silicon pillar **410a** (FIG. **6A(3)**), and then the undoped silicon pillar **610** is replaced by the highly doped silicon pillar **660a** and the third conductor pillar **630A** (FIG. **6C(3)**). Since the area of the seed region is the area of the top surface of the first highly doped silicon pillar **410a**, the area of the bottom surface of the undoped silicon pillar **610** will be the same (or substantially the same) as that of the top surface of the first highly doped silicon pillar **410a**, and the area of the bottom surface of the third conductor pillar **630A** (which replaces the undoped silicon pillar **610**) will also be the same (or substantially the same) as that of the top surface of the first highly doped silicon pillar **410a**. Thus, self-alignment is easily accomplished between the first highly doped silicon pillar **410a** and the third conductor pillar **630A**.

(89) That is, stating alternatively, as shown in FIG. **6C(3)**, a lower conductor pillar (the combination of **330A** and **410a**) has a seed region or seed pillar (such as the highly doped silicon pillar **410a**) in the upper portion thereof, and the higher conductor pillar (the combination of **630A**

and **660a**) has a seed region or seed pillar (such as the highly doped silicon pillar **660a**) in the upper portion thereof as well, the area of the top surface of the lower conductor pillar is the same or substantially the same as that of the bottom surface of the higher conductor pillar because of the self-alignment disclosed herein.

Embodiment 7

(90) According to one embodiment of the present disclosure, the method for forming a device **70** having a transistor structure **200** includes steps as follows: Step **S71**: forming a first dielectric layer over a first terminal and a second terminal of a device. Step **S72**: forming a first silicon pillar passing through the a first dielectric layer and connected to the first terminal and forming a second silicon pillar passing through the a first dielectric layer and connected to the second terminal. Step **S73**: replacing the first silicon pillar by a first conductor pillar and replacing the second silicon pillar by a second conductor pillar. Step **S74**: forming a first highly doped silicon pillar on the first conductor pillar and forming a second highly doped silicon pillar on the second conductor pillar. Step **S75**: forming a first conduction layer over the first dielectric sub-layer. Step **S76**: forming a third conductor pillar and a fourth conductor pillar respectively on the first highly doped silicon pillar and the second highly doped silicon pillar, so as to make the first conduction layer surrounds and connects to the third conductor pillar and the fourth conductor pillar. Step **S77**: forming a third highly doped silicon pillar on the third conductor pillar and forming a fourth highly doped silicon pillar on the fourth conductor pillar. Step **S78**: repeating the Steps **S75** and **S77**.

(91) The structure of the device **70** and the manufacture method thereof are similar to that of device **60**. The difference between these two is that the interconnection structure of the device **70** can further extended by repeating the steps **S75** to **S77**. Since the steps **S71** to **S77** for forming the device **70** are identical to the steps **S61** to **S67** for forming the device **60**, thus the detail procedural and material applied thereby will not redundantly described here, and the identical elements of the embodiments are designated with the same reference numerals. The process for forming the device **70** is described from step **S78**.

(92) FIG. **7A(1)** is a top view illustrating a structure after the step **S75** is repeated to form a second conduction layer **750** over the second dielectric sub-layer **640** according to one embodiment of the present disclosure. FIG. **7A(2)** is a cross-sectional view taken along the cutting line **C7A1** as depicted in FIG. **7A(1)**. FIG. **7A(3)** is a cross-sectional view taken along the cutting line **C7A2** as depicted in FIG. **7A(1)**.

(93) In the present embodiment, prior to the forming of the second conduction layer **750**, two undoped silicon pillars **710** are formed respectively on the top surface **660s** of the third highly doped silicon pillar **660a** and the fourth highly doped silicon pillar **660b**. In the present embodiment, a SEG is performed base on the top surface **660s** of the third highly doped silicon pillar **660a** and the fourth highly doped silicon pillar **660b** to grow the two undoped silicon pillars **710** respectively. Subsequently, a second dielectric layer **720** is formed to cover the first dielectric sub-layer **240**, the second dielectric sub-layer **640** and the first conduction layer **650**. Wherein each of the two undoped silicon pillars **710** has a top surface **710s** higher than the top surface **720s** of the second dielectric layer **720**.

(94) Next, the second conduction layer **750** is formed on the second dielectric layer **720** to surround and connected to these two undoped silicon pillars **710**. Subsequently, a third dielectric sub-layer **740** is formed on the top surface **750s** of the second conduction layer **750** to surround and connected to these two undoped silicon pillars **710**. Wherein, the top surface **740s** of the third dielectric sub-layer **740** substantially coplanaring with the top surface **710s** of these two undoped silicon pillars **710**.

(95) FIG. **7B(1)** is a top view illustrating a structure after a fifth conductor pillar **730A** and a sixth conductor pillar **730B** are formed on the third highly doped silicon pillar **660a** and the fourth highly doped silicon pillar **660b** according to one embodiment of the present disclosure. FIG. **7B(2)** is a cross-sectional view taken along the cutting line **C7B1** as depicted in FIG. **7B(1)**. FIG. **7B(3)** is

a cross-sectional view taken along the cutting line C7B2 as depicted in FIG. 7B(1).

(96) Similar to the step S66 for forming the third conductor pillar **630A** and the fourth conductor pillar **630B**, the fifth conductor pillar **730A** and the sixth conductor pillar **730B** are formed by replace these two undoped silicon pillars **710** with two tungsten pillars **730w** surrounded by a TiN layer **730n**.

(97) Again, similar to FIG. 6B(4) and FIG. 6B(5), besides the two undoped silicon pillars **710** are removed, the first highly doped silicon pillar **660a** and the second highly doped silicon pillar **660b** could be removed as well. Then a metal-plug process is performed to form a TiN layer **730n** and a tungsten pillars **730w** in open holes. Therefore, the fourth conductor pillar **630B** will contact the sixth conductor pillar **730B**, and the third conductor pillar **630A** will contact the fifth conductor pillar **730A**, as shown in FIG. 7B(4) and FIG. 7B(5). Thus, the resistance of this structure will be lower. The structures in FIG. 6B(4), FIG. 6B(5), FIG. 7B(4) and FIG. 7B(5) could be repeated applied to the following processes to form higher conductor pillar to other level of metal layers.

(98) FIG. 7C(1) is a top view illustrating a structure after a fifth highly doped silicon pillar **760a** and a sixth highly doped silicon pillar **760b** are formed over the fifth conductor pillar **730A** and the sixth conductor pillar **730B** as according to one embodiment of the present disclosure. FIG. 7C(2) is a cross-sectional view taken along the cutting line C7C1 as depicted in FIG. 7C(1). FIG. 7C(3) is a cross-sectional view taken along the cutting line C6C2 as depicted in FIG. 6C(1).

(99) Similar to the step S67 for forming the highly doped silicon pillar **660a** and the fourth highly doped silicon pillar **660b**, the fifth highly doped silicon pillar **760a** and the sixth highly doped silicon pillar **760b** can be formed by highly doped N+ poly silicon taking the place of the upper portions of the fifth conductor pillar **730A** and the sixth conductor pillar **730B**. Wherein the fifth highly doped silicon pillar **760a** and the sixth highly doped silicon pillar **760b** are surrounded by the second conduction layer **750** and the third dielectric sub-layer **740**, and has a top surface **760t** coplanaring with the top surface **740s** of the third dielectric sub-layer **740**.

(100) Thereinafter, the second conduction layer **750** and the third dielectric sub-layer **740** are patterned to form a third metal sub-layer **750a** and a fourth metal sub-layer **650b** serving as the connecting wires respectively surrounds and connects to the fifth conductor pillar **730A** and the sixth conductor pillar **730B**.

(101) After series steps of down-stream process are performed, the forming of the device **60** can be implemented. Again, stating alternatively, as shown in FIG. 7C(3), the previously mentioned higher conductor pillar (the combination of **630A** and **660a**) has a seed region or seed pillar (such as the highly doped silicon pillar **660a**) in the upper portion thereof, and the upper conductor pillar (the combination of **730A** and **760a**) has a seed region or seed pillar (such as the highly doped silicon pillar **760a**) in the upper portion thereof as well, the area of the top surface of the higher conductor pillar is the same or substantially the same as that of the bottom surface of the upper conductor pillar because of the self-alignment disclosed herein. Based on the FIG. 6C(3) and FIG. 7C(3), all conductor pillars could be self-aligned from the bottom conductor pillar to the upper or top conductor pillar to achieve vertical connections between them, which is different from the conventional alignment method from the upper conductor pillar to the bottom conductor pillar.

(102) While the invention has been described by way of example and in terms of the preferred embodiment (s), it is to be understood that the invention is not limited thereto. On the contrary, it is intended to cover various modifications and similar arrangements and procedures, and the scope of the appended claims therefore should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements and procedures.

(103) FIG. 8 is a cross-sectional view illustrating a device **80** having a transistor structure **200** according to further another embodiment of the present disclosure. In the present embodiment, the structure of the device **80** and the manufacture method thereof are similar to that of device **70**. The difference between these two is that the device **80** can further include other conduction layers **850a** and **850b** between the first conduction layer (e.g., the first metal sub-layer **650a**) and the upper

conduction layer (e.g., the second conduction layer **750**), wherein each of the middle conduction layers **850a** and **850b** vertically shifts with (not vertically aligns to) each other, and the conductor pillar (e.g., the fifth conductor pillar **730A**) does not electrically connect the plurality of middle conduction layers **850a** and **850b**.

Claims

1. An interconnection structure comprising: a first dielectric layer over a first conductive terminal; a first conduction layer over the first dielectric layer; a conductor pillar penetrating through the first dielectric layer and connected to the first conductive terminal; an upper dielectric layer over the first conduction layer; and an upper conduction layer over the upper dielectric layer; wherein the conductor pillar penetrates through the upper dielectric layer, the conductor pillar connects to the upper conduction layer and a seed region of the first conductive terminal, but the conductor pillar does not electrically connect the first conduction layer; the conductor pillar comprises a first epitaxial conductor pillar portion and a second epitaxial conductor pillar portion, the first epitaxial conductor pillar portion is surrounded by the first dielectric layer and the second epitaxial conductor pillar portion is surrounded by the upper dielectric layer; an area of a top surface of the first epitaxial conductor pillar portion is the same as an area of a bottom surface of the second epitaxial conductor pillar portion; a top surface of the second epitaxial conductor pillar portion is higher than a top surface of the upper dielectric layer; and the upper conduction layer contacts the top surface and sidewalls of the second epitaxial conductor pillar portion.
2. The interconnection structure according to claim 1, further comprising a plurality of middle conduction layers between the first conduction layer and the upper conduction layer, wherein each middle conduction layer vertically shifts with each other, and the conductor pillar does not electrically connect the plurality of middle conduction layers.
3. The interconnection structure according to claim 1, wherein the first epitaxial conductor pillar portion is formed on the seed region of the first conductive terminal, wherein the seed region is a silicon seed region.
4. The interconnection structure according to claim 3, wherein the seed region of the first conductive terminal is made of a poly-silicon.
5. The interconnection structure according to claim 1, wherein the second epitaxial conductor pillar portion is formed on a seed region of the first epitaxial conductor pillar portion.
6. The interconnection structure according to claim 1, wherein the first conductive terminal is a gate terminal of a semiconductor transistor; wherein the gate terminal comprises a gate dielectric layer, a gate metal layer over the gate dielectric layer, and the seed region is over the gate metal layer.
7. The interconnection structure according to claim 1, wherein the first conductive terminal is a gate terminal of a semiconductor transistor; wherein the gate terminal comprises a gate dielectric layer, and a gate poly-silicon layer over the gate dielectric layer, and the seed region is a portion of the gate poly-silicon layer.
8. The interconnection structure according to claim 1, wherein the first conductive terminal is a drain terminal of a semiconductor transistor.
9. An interconnection structure comprising: a semiconductor transistor formed on a semiconductor substrate, the semiconductor transistor having a gate terminal and a drain region; a first dielectric layer over the semiconductor transistor; a first conduction layer over the first dielectric layer; an upper dielectric layer over the first conduction layer; an upper conduction layer over the upper dielectric layer; and a conductor pillar penetrates through the first dielectric layer and the upper dielectric layer, the conductor pillar comprises a first epitaxial semiconductor pillar portion and a second epitaxial semiconductor pillar portion; a top surface of the first epitaxial semiconductor pillar portion is higher than a top surface of the first dielectric layer; and a top surface of the second

epitaxial semiconductor pillar portion is higher than a top surface of the upper dielectric layer; wherein the upper conduction layer is over the second epitaxial semiconductor pillar portion, the upper conduction layer contacts the top surface and sidewalls of the second epitaxial semiconductor pillar portion, and the first dielectric layer does not contact to the second epitaxial semiconductor pillar portion.
